

SHODWE HOTELS ANALYSIS

BUSINESS CHALLENGES :

- Hotel reports were prepared manually in Excel, making the reporting process time-consuming and prone to human errors.
- Different teams used different methods to calculate KPIs, resulting in inconsistent and unreliable business reports.
- Management did not have a single dashboard to monitor the performance of all hotels across different locations.
- There was no real-time visibility into revenue generated by individual hotels, cities, room categories, or booking platforms.
- The hotel management could not effectively track occupancy rates, making it difficult to identify underutilized properties and optimize room utilization.
- The organization lacked insights into weekday versus weekend booking patterns, preventing effective pricing and demand management.
- Hotel bookings were frequently cancelled, but there was no structured analysis to identify cancellation trends or their impact on revenue.
- Management was unable to compare hotel performance across properties, states, and room categories to identify top-performing and underperforming hotels.
- Since reports were generated manually and lacked real-time updates, business decisions were reactive instead of proactive.
- The organization had large volumes of booking data but lacked an analytical solution to transform it into meaningful insights for strategic planning and operational improvement.

PROJECT OBJECTIVE:

To design and develop an interactive Hospitality Analytics Dashboard using **Power BI** that centralizes hotel performance data, automates KPI reporting, and provides real-time insights into revenue, occupancy, booking trends, cancellations, and customer behaviour, enabling faster and data-driven business decision.

DATASET OVERVIEW:

The Hospitality Analytics project is built using a relational dataset consisting of **five interconnected tables**. The dataset captures hotel bookings, property information, room details, booking capacity, and date dimensions, enabling comprehensive analysis of hotel operations and business performance.

Dataset	Records	Description
fact_bookings	134,590	Stores detailed booking transactions, including booking status, revenue, customer information, payment method, booking platform, ratings, stay duration, and cancellation details. This serves as the primary fact table for analysis.
fact_aggregated_bookings	9,200	Contains aggregated booking information at the property and room category level, including successful bookings and room capacity, which is used to calculate Occupancy Rate and Utilization metrics.
dim_hotels	25	Provides hotel master data, including Property ID, Property Name, Hotel Category, and City for property-wise and city-wise performance analysis.
dim_rooms	4	Contains room category information, enabling analysis of revenue and occupancy by different room classes.
dim_date	92	Includes calendar attributes such as Date, Month-Year, Week Number, and Day Type (Weekday/Weekend), supporting time-based trend analysis and reporting.

DATA CLEANING & SQL MODELING

The raw hospitality datasets were cleaned, standardized, and transformed in MySQL before being connected to Power BI for analysis. The following data preparation and modeling activities were performed:

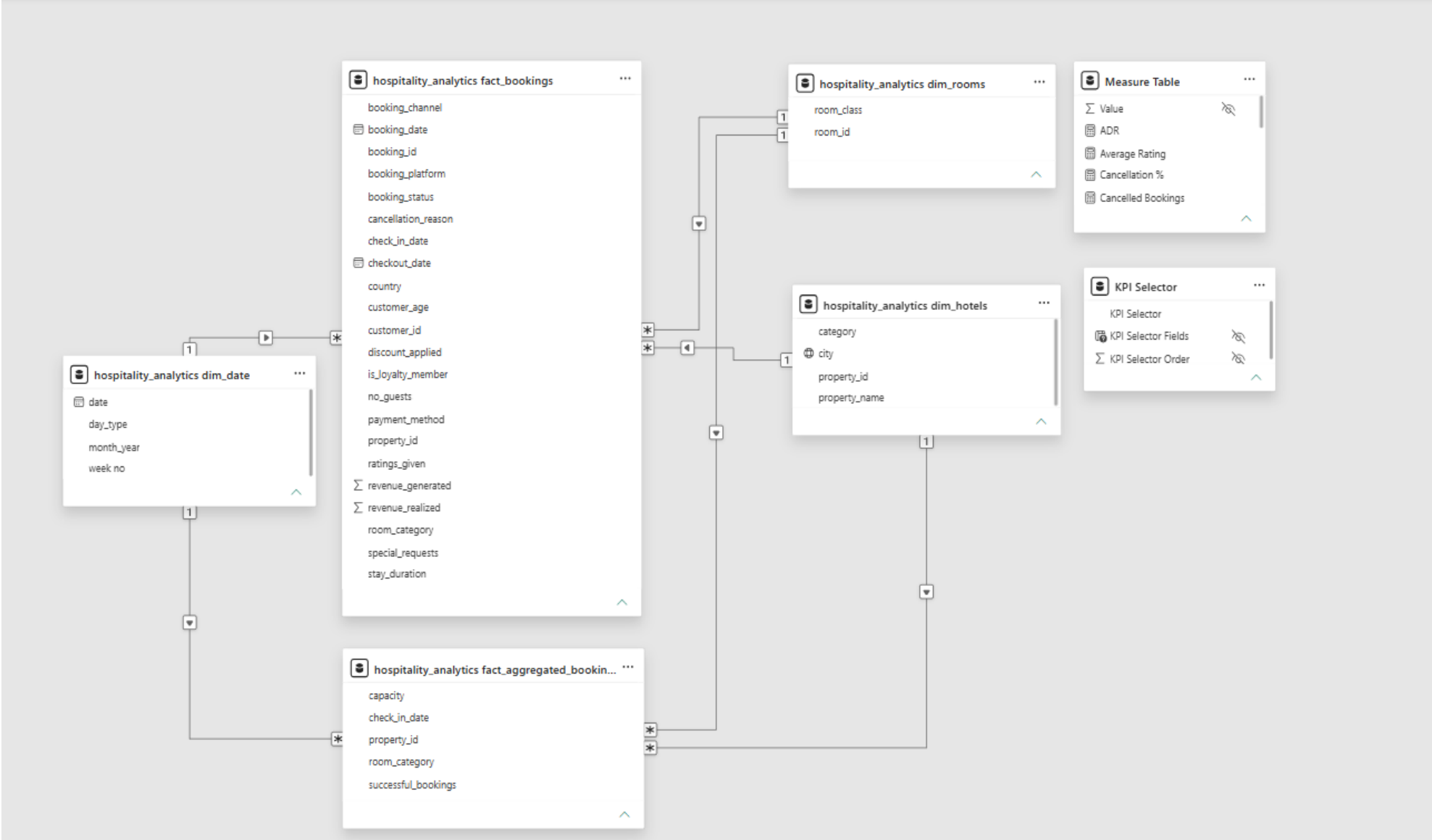
Data Cleaning

- Created a dedicated MySQL database (hospitality_analytics) to store and manage the datasets.
- Reviewed the schema of all tables to verify data types and column structures.
- Converted text-based date columns (booking_date, check_in_date, checkout_date, and date) into the SQL DATE format using STR_TO_DATE().
- Renamed incorrectly imported column names (e.g., booking_id to booking_id) to maintain consistency.
- Standardized the room_category data type to VARCHAR(10) across all relevant tables.
- Modified booking_id to VARCHAR(30) to accommodate alphanumeric booking identifiers.
- Renamed the mmm yy column to month_year for improved readability and consistency.
- Removed temporary columns created during the date conversion process after successful transformation.

SQL Data Modeling

- ❖ Assigned **Primary Keys** to:
 - dim_hotels → property_id
 - dim_rooms → room_id
 - dim_date → date
 - fact_bookings → booking_id
- ❖ Established **Foreign Key** relationships between fact and dimension tables to maintain referential integrity.
- ❖ Connected fact_bookings and fact_aggregated_bookings with:
 - dim_hotels
 - dim_rooms
 - dim_date

The project follows a **star schema** where the fact tables (fact_bookings and fact_aggregated_bookings) are connected to the dimension tables (dim_hotels, dim_rooms, and dim_date) through primary and foreign key relationships. This model improves query performance and supports efficient dashboard report



DATA VALIDATION

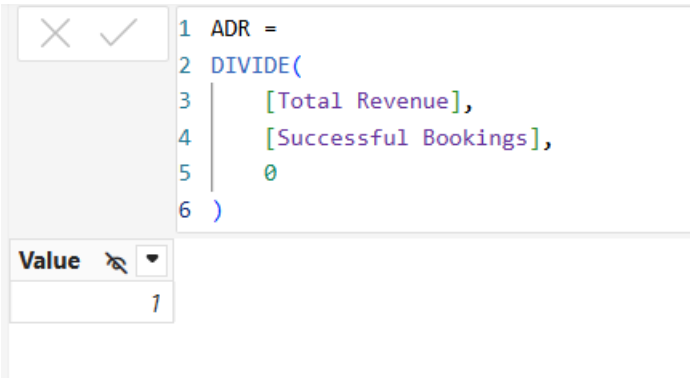
- Validated row counts after data import to ensure all records were loaded correctly.
- Performed duplicate checks on booking_id to confirm record uniqueness.
- Verified that key fields such as property_id, room_category, and booking_id contained no null values.
- Checked for invalid date records, ensuring that checkout_date was always later than check_in_date.
- Validated all primary and foreign key relationships to ensure data integrity before connecting the database to Power BI.

DAX MEASURES & KPI CALCULATIONS

To support dynamic reporting and real-time business analysis, multiple DAX (Data Analysis Expressions) measures were created in Power BI. These measures were used to calculate key hospitality KPIs, perform revenue analysis, monitor occupancy trends, evaluate booking performance, and support interactive dashboard visualizations.

The following DAX measures were implemented in the project:

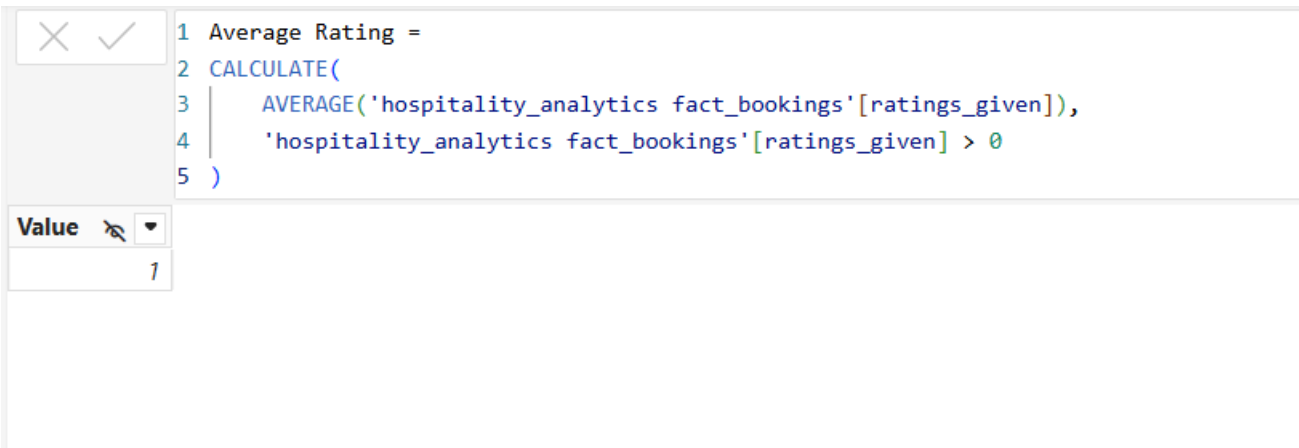
ADR:



Calculates the average revenue earned per occupied room.

Business Use Case:
Helps assess pricing strategy effectiveness, measure customer spending behaviour, and identify opportunities to maximize revenue without reducing room prices.

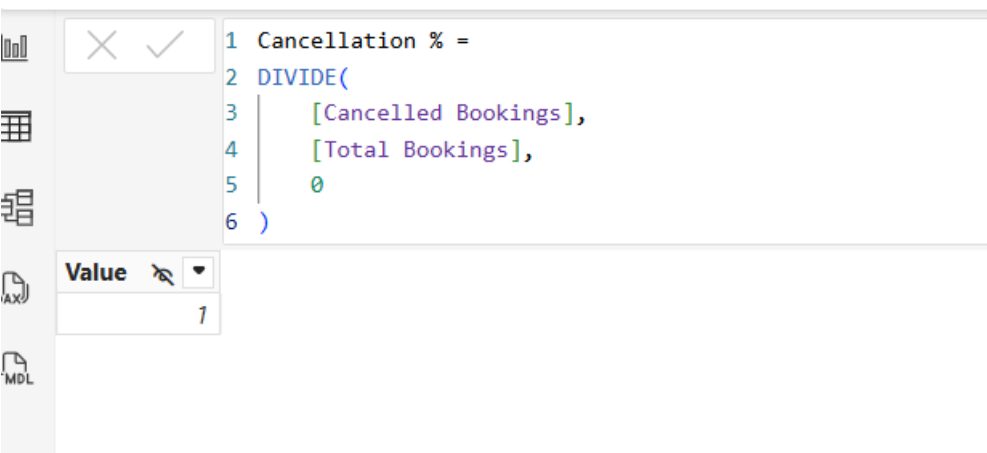
AVERAGE CUSTOMER RATING:



Calculates the average customer rating provided by hotel guests.

Business Use Case:
Used to measure customer satisfaction, evaluate service quality, identify areas for operational improvement, and monitor guest experience across properties.

CANCELLATION RATE:



Calculates the percentage of bookings that were cancelled out of total bookings.

Business Use Case:
Helps management identify cancellation trends, evaluate revenue leakage, improve forecasting accuracy, and implement strategies to reduce booking cancellations.

OCCUPANCY RATE:

✕

✓

Value 

1

```
1 Occupancy % =
2 DIVIDE(
3   [Successful Bookings],
4   [Total Capacity],
5   0
6 )
```

Calculates the percentage of occupied rooms compared to the total available room capacity.

Business Use Case:
Used to evaluate room utilization efficiency, identify underutilized properties, optimize inventory management, and improve occupancy performance.

REVPAR:

✕

✓

Value 

1

```
1 RevPAR =
2 DIVIDE(
3   [Total Revenue],
4   [Total Capacity],
5   0
6 )
```

Measures the revenue generated per available room by combining occupancy and pricing performance.

Business Use Case:
Used to evaluate overall hotel performance, compare property efficiency, and identify hotels that effectively balance room occupancy with pricing strategy.

SUCCESSFUL BOOKINGS:

✕

✓

Value 

1

```
1 Successful Bookings =
2 SUM('hospitality_analytics fact_aggregated_bookings'[successful_bookings])
```

Calculates the total number of successfully completed bookings.

Business Use Case:
Used to monitor actual business conversions, evaluate booking efficiency, and compare successful bookings across properties and booking platforms.

TOTAL BOOKINGS:

✕

✓

Value 

1

```
1 Total Bookings =
2 DISTINCTCOUNT('hospitality_analytics fact_bookings'[booking_id])
```

Counts the total number of bookings made across all hotel properties.

Business Use Case:
Helps analyze customer demand patterns, booking trends, seasonal variations, and overall hotel activity levels.

TOTAL REVENUE:

✕ ✓ 1 Total Revenue = 2 SUM('hospitality_analytics fact_bookings'[revenue_realized]))
Value 1

Calculates the total revenue generated from all hotel bookings recorded in the dataset.

Business Use Case:

Used by management to monitor overall financial performance, compare revenue across hotels and cities, identify high-performing properties, and evaluate business growth trends.

OTHER DAX MEASURES UTILIZED:

✕ ✓ 1 Cancelled Bookings = 2 CALCULATE(3 DISTINCTCOUNT('hospitality_analytics fact_bookings'[booking_id]), 4 'hospitality_analytics fact_bookings'[booking_status] = "Cancelled" 5)
Value 1

✕ ✓ 1 Checked Out Bookings = 2 CALCULATE(3 DISTINCTCOUNT('hospitality_analytics fact_bookings'[booking_id]), 4 'hospitality_analytics fact_bookings'[booking_status] = "Checked Out" 5)
Value 1

✕ ✓ 1 Total Capacity = 2 SUM('hospitality_analytics fact_aggregated_bookings'[capacity]))
Value 1

✕ ✓ 1 Revenue Contribution % = 2 DIVIDE(3 [Total Revenue], 4 CALCULATE(5 [Total Revenue], 6 ALL('hospitality_analytics dim_rooms') 7) 8)
Value 1

✕ ✓ 1 Revenue Generated = 2 SUM('hospitality_analytics fact_bookings'[revenue_generated]))
Value 1

✕

✓

1 Selected Property =
2 SELECTEDVALUE('hospitality_analytics dim_hotels'[property_name])

Value

1

✕

✓

1 KPI Selector = {
2 ("Total Revenue", NAMEOF('Measure Table'[Total Revenue]), 0),
3 ("Occupancy %", NAMEOF('Measure Table'[Occupancy %]), 1),
4 ("ADR", NAMEOF('Measure Table'[ADR]), 2),
5 ("RevPAR", NAMEOF('Measure Table'[RevPAR]), 3),
6 ("Cancellation %", NAMEOF('Measure Table'[Cancellation %]), 4),
7 ("Total Bookings", NAMEOF('Measure Table'[Total Bookings]), 5)
8 }

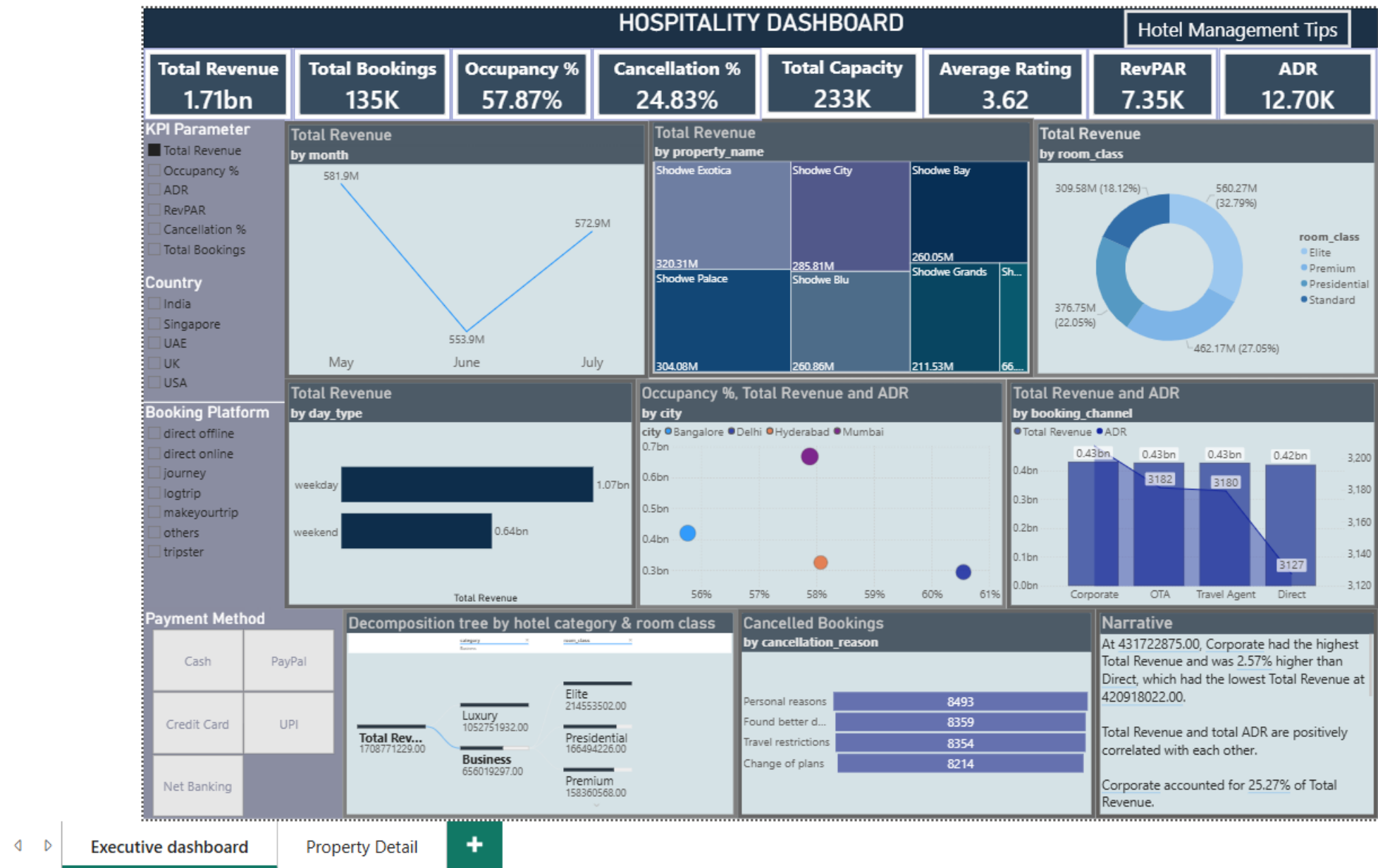
KPI Selector	KPI Selector Fields	KPI Selector Order
Total Revenue	'Measure Table'[Total Revenue]	0
Occupancy %	'Measure Table'[Occupancy %]	1
ADR	'Measure Table'[ADR]	2
RevPAR	'Measure Table'[RevPAR]	3
Cancellation %	'Measure Table'[Cancellation %]	4
Total Bookings	'Measure Table'[Total Bookings]	5

DASHBOARD WALKTHROUGH WITH KEY INSIGHTS

ADR(Average daily rate)- Average rental income per paid occupied room

RevPAR(Revenue generated per available room)- Determines the effectiveness of a property that can fill its rooms while balancing its average price charged depends on occupancy and pricing performance

REVENUE ANALYSIS:



- KPIs indicate organization generated a total revenue of 1.71 billion from 135 thousand bookings. Occupancy rate stands at 57.87% which means nearly 42% of available rooms are not being utilized. Average daily rate is 12.7 thousand indicating strong pricing power. However RevPAR is significantly lower than ADR, indicating that a considerable number of rooms remain unoccupied. Cancellation rate is 24.83% means approximately 1 out of 4 rooms booked gets cancelled and average customer rating is 3.62 out of 5 highlighting room for improvement in guest satisfaction.
- Revenue trend reveals fluctuations across the three months. The decline in June may suggest possible issues such as reduce in demand, higher cancellations or lower occupancy.
- Mumbai is strongest market generating close to 700 million revenue. Delhi shows relatively higher occupancy but lower revenue compared to Mumbai.
- Weekday revenue exceeds weekend revenue, indicating stronger corporate and business travel demand
- Room class analysis demonstrates that guests are willing to spend more for premium experiences. An effective strategy would be to encourage room upgrades during booking process and create premium packages to increase average booking value.

OCCUPANCY ANALYSIS:



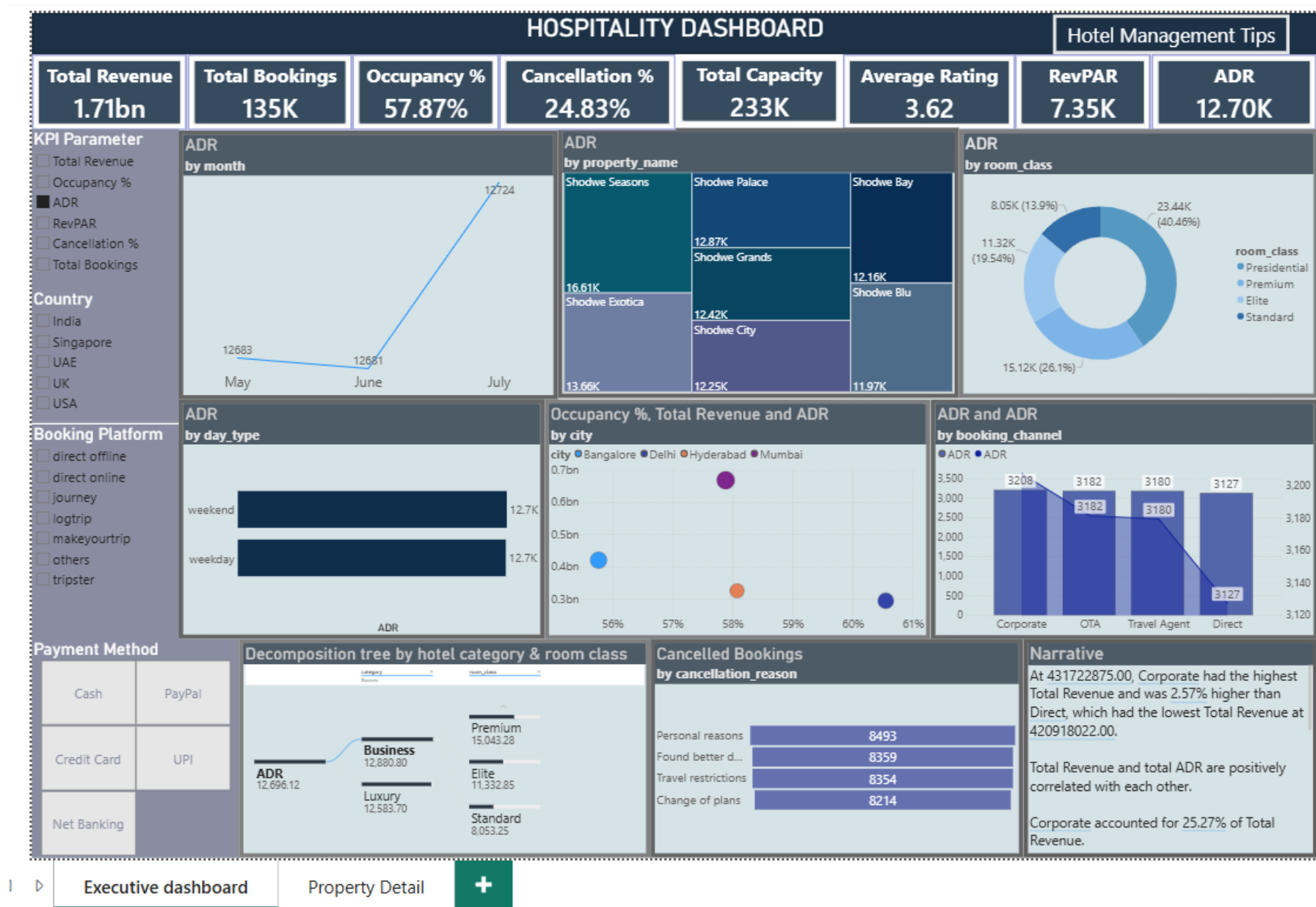
Executive dashboard

Property Detail



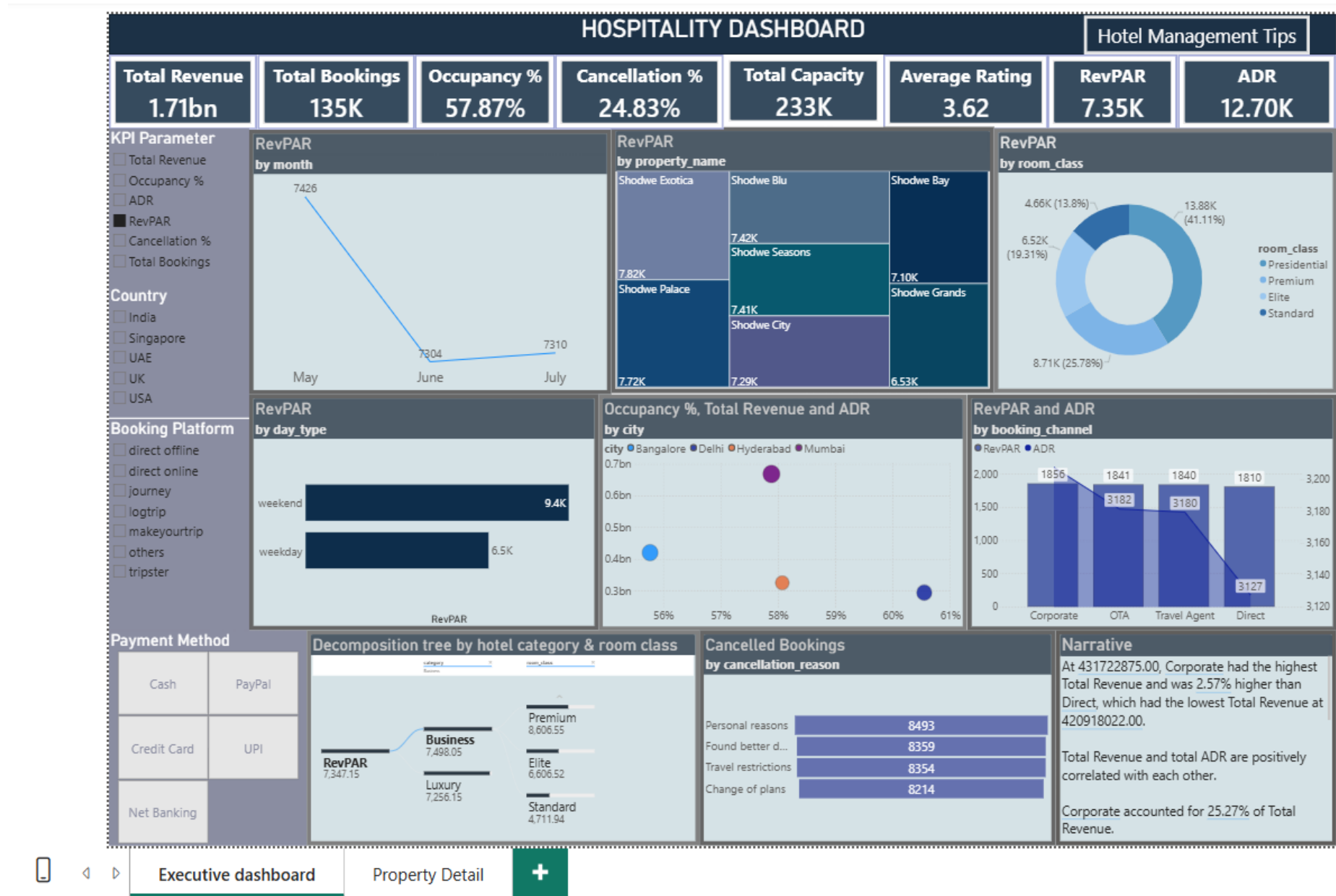
- Occupancy is higher on weekends at 73.58% while weekdays have 51.34% indicating leisure demand while
- weekdays benefit from higher room rate since revenue is more on weekdays.
- Decomposition tree suggests that demand is distributed fairly across room categories suggesting occupancy challenges are affecting all room categories rather than a specific segment.
- Through targeted promotions, loyalty programs or corporate partnerships more guests can be attracted without investing in additional infrastructure.

ADR ANALYSIS:



- Corporate bookings deliver highest ADR while direct bookings generate lowest ADR deriving fact that corporate customers contribute higher value per booking.
- Presidential rooms achieve highest ADR of 23.44k almost 3 times ADR of standard rooms.
- It indicates that customers are willing to pay premium prices for hotel offerings. Therefore reducing room rates should not be primary strategy for growth. Instead organization should focus on increasing occupancy while maintaining current pricing levels.

RevPAR ANALYSIS:



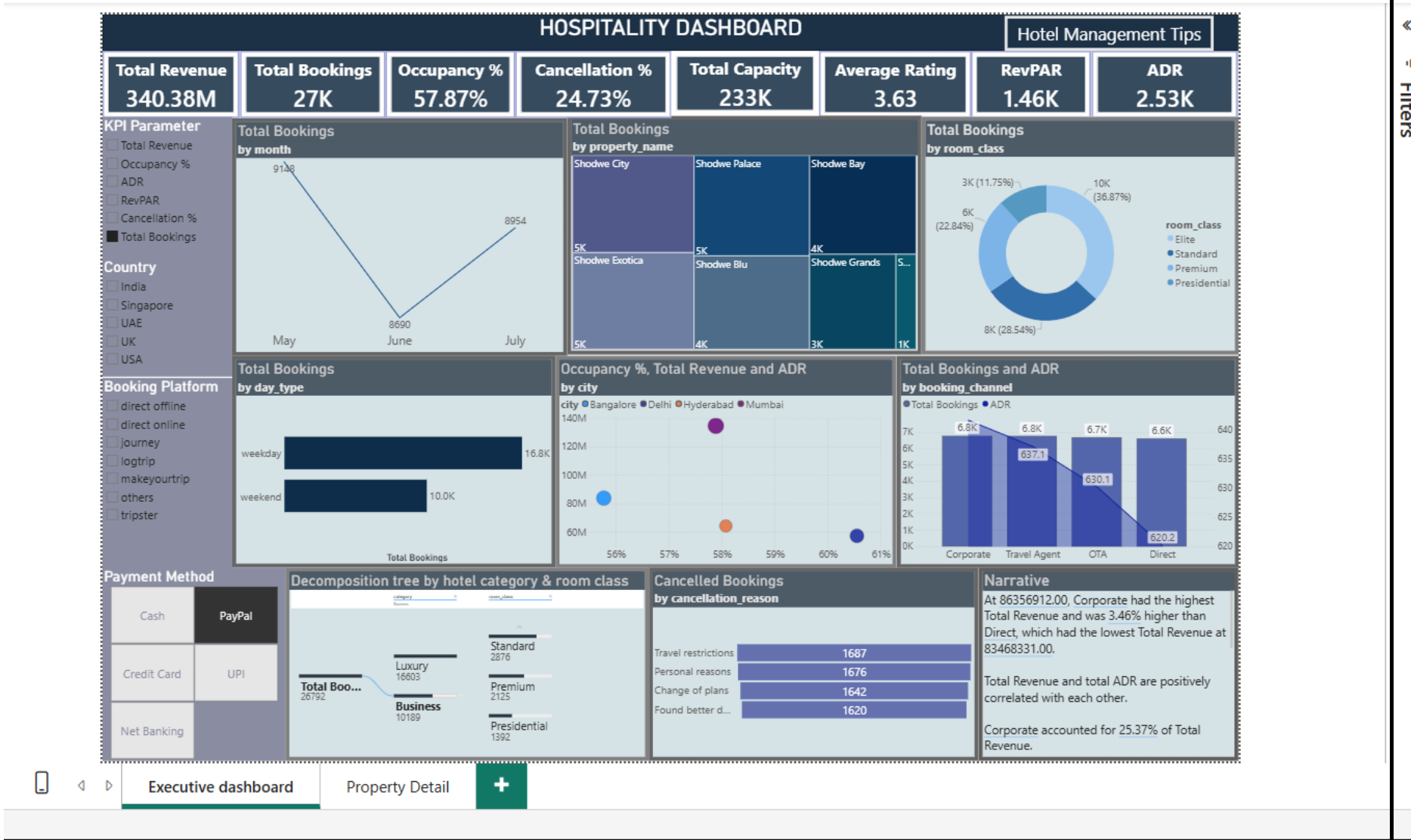
- Among all properties Shodwe Exotica performs best with RevPAR of 7.82 thousand. Shodwe Grands records lowest RevPAR of 6.53 thousand.
- This performance gap indicates opportunities to learn from top-performing properties and replicate their pricing, marketing and customer service practices across lower performing locations.
- Weekend RevPAR is higher than weekday RevPAR indicating weaker weekday demand.
- To address this gap organization can introduce corporate travel packages, business traveller promotions and work from hotel offering.

CANCELLATION ANALYSIS:



- Personal reasons are leading cause of cancellation with 8493 cases & price competition is other major issue with over 8300 customers cancelling after finding better deals elsewhere.
- High cancellation rate creates revenue uncertainty reducing forecasting accuracy and impacts room inventory planning.
- Solutions include stricter cancellation policy, advance payment requirements & predictive models to anticipate cancellation behaviour.

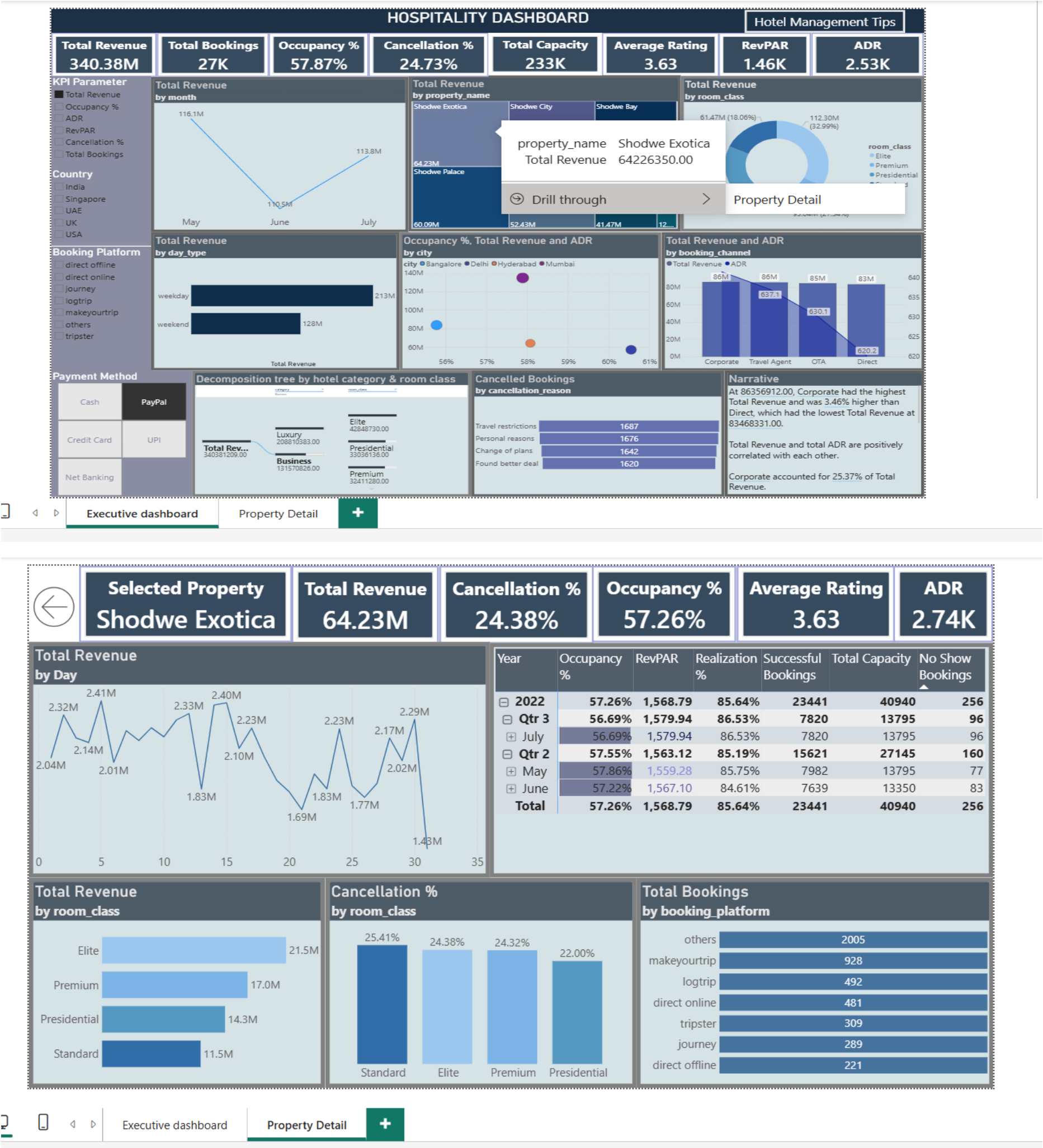
BOOKINGS ANALYSIS:



- Online travel agencies & corporate channels contribute highest booking volume while these channels help generate demand they involve commission costs.
- Increasing direct bookings through loyalty programs, website promotions and targeted marketing campaigns can improve profitability by reducing dependency on third party platforms.

DASHBOARD FEATURES

DRILL THROUGH:



- Using the Treemap on the Executive Dashboard, a drill-through action was performed to navigate to the Property Detail page for *Shodwe Exotica* for detailed performance analysis.
- The drill-through revealed that Shodwe Exotica generated ₹64.23M in revenue with a 57.26% occupancy rate and a 24.38% cancellation rate, making it one of the top-performing properties while highlighting room for improvement in booking retention.
- Further analysis showed that Elite rooms contributed the highest revenue (₹21.5M), and most bookings were received through third-party platforms, indicating opportunities to increase direct bookings and improve profitability.

FILTERS & NARRATIVE



The dashboard includes multiple interactive filters and slicers to enable dynamic analysis and improve user-driven exploration of hotel performance data. These filters allow management to analyze KPIs across different business dimensions and make faster data-driven decisions.

- KPI Parameter Filter**
Allows users to dynamically switch between KPIs such as Total Revenue, Occupancy %, ADR, RevPAR, Cancellation %, and Total Bookings.
- Country Filter**
Enables analysis of hotel performance across different countries including India, Singapore, UAE, UK, and USA.
- Booking Platform Filter**
Allows users to analyze booking performance based on different booking channels such as direct online, direct offline, OTA platforms, and travel agencies.
- Payment Method Filter**
Filters booking transactions based on payment methods including Cash, PayPal, Credit Card, UPI, and Net Banking.

NARRATIVE: Used to automatically generate data-driven business summaries and highlight key insights, enabling management to quickly understand dashboard performance and make faster strategic decisions.

CONCLUSION

The Hospitality Analytics Dashboard successfully transformed raw hotel booking data into meaningful business insights through interactive visualizations, KPI tracking, SQL data modeling, and DAX-based calculations. The analysis identified important trends related to revenue performance, occupancy patterns, booking behaviour, cancellations, and customer preferences, enabling better operational and strategic decision-making. By providing real-time and centralized reporting, the dashboard helps hotel management improve pricing strategies, optimize room utilization, reduce revenue leakage, and support data-driven business growth.